

基于 Hu–Washizu 变分原理的薄壳结构 准变分一致无网格法

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第五届无网格粒子类方法进展与应用研讨会 山西·太原

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提纲

1. 薄壳结构伽辽金无网格法
2. 薄壳问题变分一致性与准变分一致积分方案
3. 基于 Hu-Washizu 原理变分一致型本质边界条件施加方法
4. 结论

薄壳结构

壳体中面应变与曲率

$$\begin{aligned}\epsilon_{\alpha\beta} &= \frac{1}{2}(\mathbf{a}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{a}_\beta) \\ &= \epsilon_{\alpha\beta} + \kappa_{\alpha\beta} \xi^3\end{aligned}$$

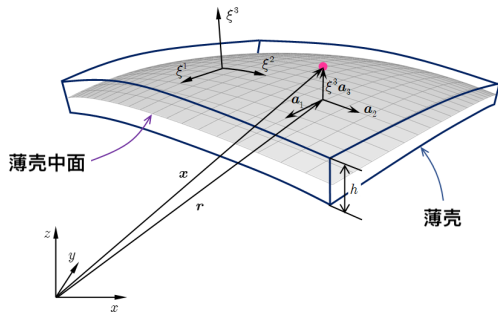
Kirchhoff 假设

$$\epsilon_{3i} = 0 \Rightarrow \theta = -\mathbf{u}_{,\alpha} \cdot \mathbf{a}_3 \mathbf{a}^\alpha$$



薄壳结构曲率

$$\begin{aligned}\epsilon_{\alpha\beta} &= \frac{1}{2}(\mathbf{a}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{a}_\beta) \\ \kappa_{\alpha\beta} &= (\Gamma_{\alpha\beta}^\gamma \mathbf{u}_{,\gamma} - \mathbf{u}_{,\alpha\beta}) \cdot \mathbf{a}_3 = -\mathbf{u}_{,\alpha|\beta} \cdot \mathbf{a}_3\end{aligned}$$



薄壳结构中面

薄壳结构分析需要 C_1 连续形函数

再生核无网格近似

位移近似

$$u^h(\xi) = \sum_{I=1}^{n_p} \Psi_I(\xi) d_I$$

$$\Psi_I(\xi) = p^T(\xi) c(\xi) \phi(\xi_I - \xi)$$

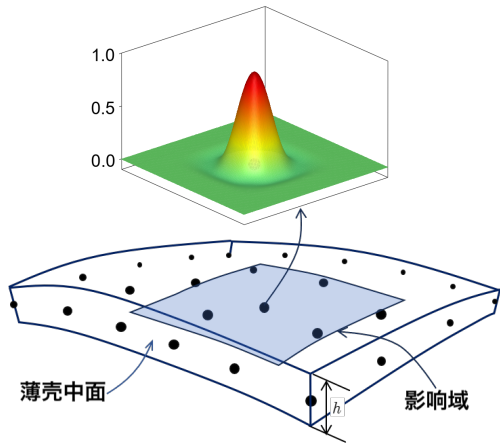
一致性条件

$$\sum_{I=1}^{n_p} \Psi_I(\xi) p(\xi_I) = p(\xi)$$

$$\Rightarrow c(\xi) = A^{-1}(\xi) p(0)$$

无网格形函数

$$\Psi_I(\xi) = p^T(\xi_I - \xi) A^{-1}(\xi) p(0) \phi(\xi_I - \xi)$$

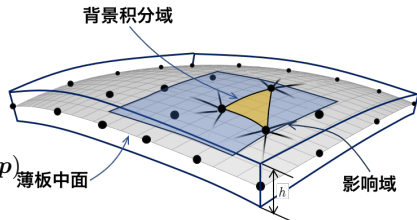


再生核无网格近似

变分一致性

变分一致性阶次等于伽辽金弱形式能准确求解由多项式所组成的精确解阶次:

$$\int_{\Omega} h \delta \varepsilon_{\alpha\beta} C^{\alpha\beta\gamma\eta} \varepsilon_{\gamma\eta}(\mathbf{p}) d\Omega + \int_{\Omega} \frac{h^3}{12} \delta \kappa_{\alpha\beta} C^{\alpha\beta\gamma\eta} \kappa_{\gamma\eta}(\mathbf{p}) d\Omega = f(\mathbf{p})$$



伽辽金无网格法数值积分

伽辽金无网格法误差估计 (Wu and Wang, 2021)

变分一致性条件控制

$$\|\mathbf{u} - \mathbf{u}_h\|_{H_e} \leq \boxed{C_p s^p |\mathbf{u}|_{H_p}} + \boxed{C_q s^q |\mathbf{u}|_{H_q}}$$

一致性条件控制

变分一致型边界条件施加方案

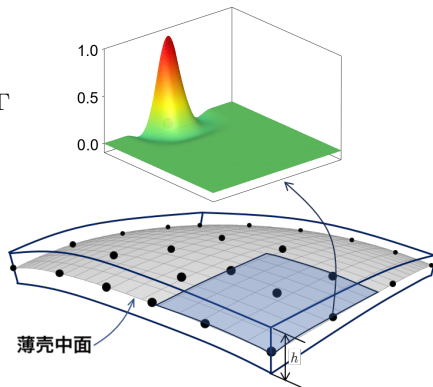
Nitsche 法施加边界条件

- 稳定项

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta a_3 a_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I a_3 a_3 \Psi_J]]_{x \in C_v}$$

- 一致项

$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma a_3 M_{nnJ} + a_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I a_3 \mathbf{P}_J]] + [[\mathbf{P}_I a_3 \Psi_J]])_{x \in C_v}$$



边界点处无网格形函数

变分一致型边界条件施加方案

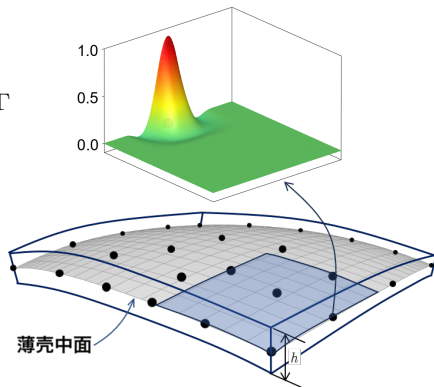
Nitsche 法施加边界条件

- 稳定项 $\times$ 人工经验参数影响求解精度

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta a_3 a_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I a_3 a_3 \Psi_J]]_{x \in C_v}$$

- 一致项

$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma a_3 M_{nnJ} + a_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I a_3 \mathbf{P}_J]] + [[\mathbf{P}_I a_3 \Psi_J]])_{x \in C_v}$$



边界点处无网格形函数

变分一致型边界条件施加方案

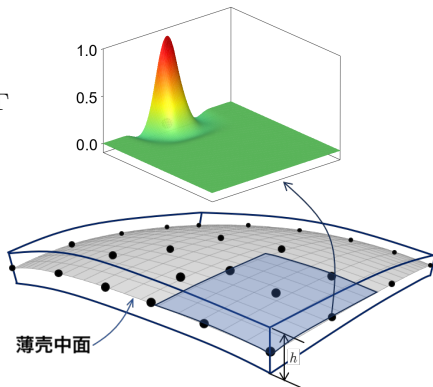
Nitsche 法施加边界条件

- 稳定项 × 人工经验参数影响求解精度

$$K_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta a_3 a_3 n^\gamma \Psi_{J,\gamma} d\Gamma \\ + \alpha_C [[\Psi_I a_3 a_3 \Psi_J]]_{x \in C_v}$$

- 一致项 × 需计算形函数高阶导数

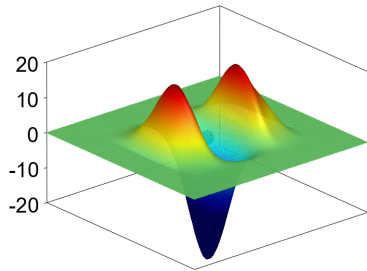
$$K_{IJ}^c = - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma a_3 M_{nnJ} + a_3 M_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ + ([[\Psi_I a_3 \mathbf{P}_J]] + [[\mathbf{P}_I a_3 \Psi_J]])_{x \in C_v}$$



边界点处无网格形函数

无网格形函数高阶导数

$$\Psi_{I,ij}(\xi) = \begin{pmatrix} p_{,ij}^T(\xi_I - \xi)A^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,i}^T(\xi_I - \xi)A_{,j}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,i}^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,j}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,ij}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p_{,j}^T(\xi_I - \xi)A_{,i}^{-1}(\xi)\phi_s(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,i}^{-1}(\xi)\phi_{s,j}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,ij}(\xi_I - \xi) \\ +p_{,j}^T(\xi_I - \xi)A^{-1}(\xi)\phi_{s,i}(\xi_I - \xi) \\ +p^T(\xi_I - \xi)A_{,j}^{-1}(\xi)\phi_{s,i}(\xi_I - \xi) \end{pmatrix} p(0)$$



形函数二阶导数

无网格形函数高阶导数
计算复杂耗时

基于 Hellinger-Reissner 变分一致型伽辽金无网格法

薄板问题 Hellinger-Reissner 变分原理弱形式

$$\begin{aligned} \int_{\Omega} \delta M_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} M_{\gamma\eta} d\Omega + \int_{\Gamma_V} \delta V_n w d\Gamma - \int_{\Gamma_M} \delta M_{nn} w_{,n} d\Gamma &= \int_{\Gamma_w} \delta V_n \bar{w} d\Gamma - \int_{\Gamma_{\theta}} \delta M_{nn} \bar{\theta}_n d\Gamma \\ \int_{\Gamma_V} \delta w V_n d\Gamma - \int_{\Gamma_M} \delta w_{,n} M_{nn} d\Gamma &= \int_{\Gamma_V} \delta w \bar{V}_n d\Gamma - \int_{\Gamma_M} \delta w_{,n} \bar{M}_{nn} d\Gamma + \int_{\Omega} \delta w \bar{q} d\Omega \end{aligned}$$

无网格近似挠度

$$w(\boldsymbol{x}) = \sum_{I=1}^{n_p} \Psi_I(\boldsymbol{x}) d_I^w$$

局部多项式近似弯矩

$$M_{\alpha\beta}(\boldsymbol{x}) = \boldsymbol{q}^T(\boldsymbol{x}) \boldsymbol{d}_{\alpha\beta}^M \quad \text{in } \Omega_C$$

满足全域变分一致性，避免计算复杂耗时的形函数高阶导数

基于 Hellinger-Reissner 变分一致型伽辽金无网格法

薄板问题 Hellinger-Reissner 变分原理弱形式

$$\begin{aligned} \int_{\Omega} \delta M_{\alpha\beta} D_{\alpha\beta\gamma\eta}^{-1} M_{\gamma\eta} d\Omega + \int_{\Gamma_V} \delta V_n w d\Gamma - \int_{\Gamma_M} \delta M_{nn} w_{,n} d\Gamma &= \int_{\Gamma_w} \delta V_n \bar{w} d\Gamma - \int_{\Gamma_\theta} \delta M_{nn} \bar{\theta}_n d\Gamma \\ \int_{\Gamma_V} \delta w V_n d\Gamma - \int_{\Gamma_M} \delta w_{,n} M_{nn} d\Gamma &= \int_{\Gamma_V} \delta w \bar{V}_n d\Gamma - \int_{\Gamma_M} \delta w_{,n} \bar{M}_{nn} d\Gamma + \int_{\Omega} \delta w \bar{q} d\Omega \end{aligned}$$

无网格近似挠度

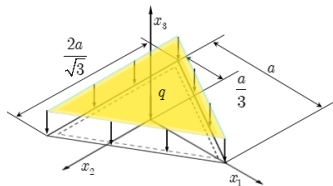
$$w(\mathbf{x}) = \sum_{I=1}^{n_p} \Psi_I(\mathbf{x}) d_I^w$$

局部多项式近似弯矩

$$M_{\alpha\beta}(\mathbf{x}) = \mathbf{q}^T(\mathbf{x}) \mathbf{d}_{\alpha\beta}^M \quad \text{in } \Omega_C$$

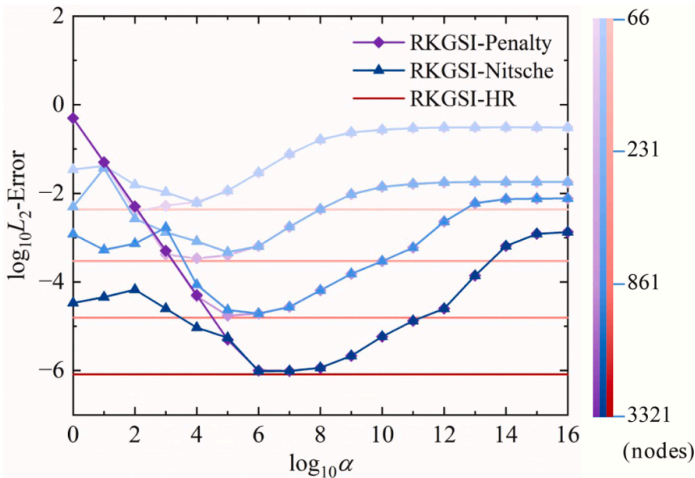
Hellinger-Reissner 变分原理弱形式包含本质边界条件施加过程，
无需人工经验参数

基于 Hellinger-Reissner 变分一致型伽辽金无网格法



简支三角形板问题

抗弯刚度 $D = 1.0$
泊松比 $\nu = 0.3$
荷载 $q = 1.0$



人工经验参数敏感性分析

薄壳伽辽金法的变分一致性

假定位移为由基函数构成的多项式

$$\begin{array}{ccccc} u(\xi) = p^T(\xi)a & \Rightarrow & N^{\alpha\beta} = hC^{\alpha\beta\gamma\eta}\frac{1}{2}(\mathbf{a}_\gamma \cdot \mathbf{u}_{,\eta} + \mathbf{u}_{,\gamma} \cdot \mathbf{a}_\eta) & \Rightarrow & \mathbf{T}_N = \mathbf{a}_\alpha N^{\alpha\beta} n_\beta \\ \text{多项式} & & \text{非多项式} & & \text{非多项式} \end{array}$$

薄壳薄膜应变部分变分一致性条件

$$\int_{\Omega} \delta \varepsilon_{\alpha\beta} N^{\alpha\beta} d\Omega = \int_{\Gamma} \delta \mathbf{u} \cdot \mathbf{T}_N d\Gamma - \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{b}_N d\Omega$$

多项式一致性条件 $\Leftrightarrow$ 非多项式变分一致性条件:

- 位移为多项式，可能需要满足非多项式变分一致性条件。
- 假设应力为多项式，变分一致性条件的右端也存在非线性项。

薄壳问题的 Hu-Washizu 变分原理伽辽金弱形式

薄壳薄膜应变部分变分一致性条件

$$\int_{\Omega} \underbrace{\delta \varepsilon_{\alpha\beta} N^{\alpha\beta}}_{\frac{1}{2}(\mathbf{a}_{\alpha} N^{\alpha\beta} \cdot \delta \mathbf{u}_{,\beta} + \delta \mathbf{u}_{,\alpha} \cdot \mathbf{a}_{\beta} N^{\alpha\beta})} d\Omega = \int_{\Gamma} \delta \mathbf{u} \cdot \underbrace{\mathbf{T}_N}_{\mathbf{a}_{\alpha} N^{\alpha\beta} n_{\beta}} d\Gamma - \int_{\Omega} \delta \mathbf{u} \cdot \underbrace{\mathbf{b}_N}_{(\mathbf{a}_{\alpha} N^{\alpha\beta})|_{\beta}} d\Omega$$

- 假定应变: $\mathbf{u}_{,\alpha}(\xi) = \mathbf{q}^T(\xi) \mathbf{d}_{\alpha}^{\varepsilon} \Rightarrow \varepsilon_{\alpha\beta}(\xi) = \mathbf{q}^T(\xi) \frac{1}{2}(\mathbf{a}_{\alpha} \cdot \mathbf{d}_{\beta}^{\varepsilon} + \mathbf{a}_{\beta} \cdot \mathbf{d}_{\alpha}^{\varepsilon})$
- 假定应力: $\mathbf{a}_{\alpha} N^{\alpha\beta}(\xi) = \mathbf{q}^T(\xi) \mathbf{d}_N^{\beta} \Rightarrow N^{\alpha\beta}(\xi) = \mathbf{q}^T(\xi) \mathbf{a}^{\alpha} \cdot \mathbf{d}_N^{\beta}$

薄膜应变部分 Hu-Washizu 变分原理

$$-\int_{\Omega} h \delta \varepsilon_{\alpha\beta} C^{\alpha\beta\gamma\eta} \varepsilon_{\gamma\eta} d\Omega + \int_{\Omega} \delta \varepsilon_{\alpha\beta} N^{\alpha\beta} d\Omega = 0$$
$$\int_{\Omega} \delta N^{\alpha\beta} \varepsilon_{\alpha\beta} d\Omega - \int_{\Gamma} \delta \mathbf{T}_N \cdot \mathbf{u} d\Gamma + \int_{\Omega} \delta \mathbf{b}_N \cdot \mathbf{v} d\Omega + \int_{\Gamma_u} \delta \mathbf{T}_N \cdot \mathbf{u} d\Gamma = \int_{\Gamma_u} \delta \mathbf{T}_N \cdot \bar{\mathbf{u}} d\Gamma$$

薄壳问题的 Hu-Washizu 变分原理伽辽金弱形式

薄壳弯曲应变部分变分一致性条件

$$\int_{\Omega} \delta \kappa_{\alpha\beta} M^{\alpha\beta} d\Omega = \int_{\Gamma} \delta \theta_n M_{nn} d\Gamma - \int_{\Gamma} \delta \mathbf{u} \cdot \mathbf{T}_M d\Gamma - (\delta \mathbf{u} \cdot P \mathbf{a}_3)_{x \in C} - \int_{\Omega} \mathbf{u} \cdot \mathbf{b}_M d\Omega$$

$$\delta \kappa_{\alpha\beta} M^{\alpha\beta} = -\delta \mathbf{u}_{,\alpha|\beta} \cdot \mathbf{a}_3 M^{\alpha\beta}$$

$$\delta \theta_n M_{nn} = -\delta \mathbf{u}_{,\alpha} \cdot \mathbf{a}_3 M^{\beta\gamma} n^\alpha n_\beta n_\gamma$$

$$\mathbf{T}_M = (\mathbf{a}_3 M^{\alpha\beta} s_\alpha n_\beta)_{,\gamma} s^\gamma + (\mathbf{a}_3 M^{\alpha\beta})|_\beta n_\alpha$$

$$P \mathbf{a}_3 = -[(\mathbf{a}_3 M^{\alpha\beta} s_\alpha n_\beta)]$$

$$\mathbf{b}_M = (\mathbf{a}_3 M^{\alpha\beta})|_{\alpha\beta}$$



- 假定应变: $-\mathbf{u}_{,\alpha|\beta}(\xi) = \mathbf{q}^T(\xi) d_{\alpha\beta}^\kappa \Rightarrow \kappa_{\alpha\beta}(\xi) = \mathbf{q}^T(\xi) \mathbf{a}_3 \cdot d_{\alpha\beta}^\kappa$
- 假定应力: $\mathbf{a}_3 M^{\alpha\beta h}(\xi) = \mathbf{q}^T(\xi) d_M^{\alpha\beta} \Rightarrow M^{\alpha\beta}(\xi) = \mathbf{q}^T(\xi) \mathbf{a}_3 \cdot d_M^{\alpha\beta}$

准变分一致型伽辽金无网格法

位移离散 ✓ 保证一致性

$$u^h(\xi) = \sum_{I=1}^{n_p} \Psi_I(\xi) d_I$$

$$p(\xi) = \{1, \xi^1, \xi^2, (\xi^1)^2, \xi^1 \xi^2, (\xi^2)^2\}^T$$

应变与应力离散 ✓ 保证变分一致性

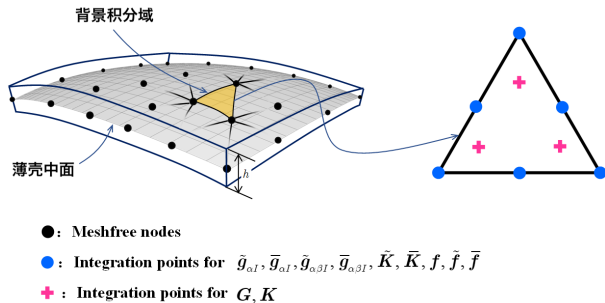
$$\varepsilon_{\alpha\beta}(\xi) = q^T(\xi) \frac{1}{2} (a_\alpha \cdot d_\beta^\varepsilon + a_\beta \cdot d_\alpha^\varepsilon)$$

$$\kappa_{\alpha\beta}(\xi) = q^T(\xi) a_3 \cdot d_{\alpha\beta}^\kappa$$

$$N^{\alpha\beta}(\xi) = q^T(\xi) a^\alpha \cdot d_N^\beta$$

$$M^{\alpha\beta}(\xi) = q^T(\xi) a_3 \cdot d_M^{\alpha\beta}$$

$$q(\xi) = \{1, \xi^1, \xi^2\}^T$$



准变分一致型伽辽金无网格法

位移离散 ✓ 保证一致性

$$\mathbf{u}^h(\boldsymbol{\xi}) = \sum_{I=1}^{n_p} \Psi_I(\boldsymbol{\xi}) \mathbf{d}_I$$

$$\mathbf{p}(\boldsymbol{\xi}) = \{1, \xi^1, \xi^2, (\xi^1)^2, \xi^1 \xi^2, (\xi^2)^2\}^T$$

应变与应力离散 ✓ 保证变分一致性

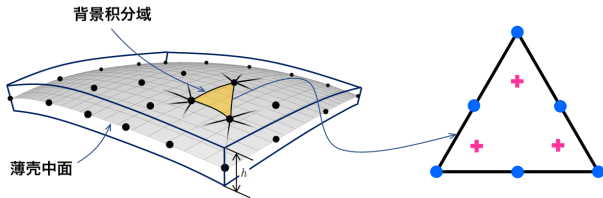
$$\varepsilon_{\alpha\beta}(\boldsymbol{\xi}) = \mathbf{q}^T(\boldsymbol{\xi}) \frac{1}{2} (\mathbf{a}_\alpha \cdot \mathbf{d}_{\beta}^{\varepsilon} + \mathbf{a}_\beta \cdot \mathbf{d}_{\alpha}^{\varepsilon})$$

$$\kappa_{\alpha\beta}(\boldsymbol{\xi}) = \mathbf{q}^T(\boldsymbol{\xi}) \mathbf{a}_3 \cdot \mathbf{d}_{\alpha\beta}^{\kappa}$$

$$N^{\alpha\beta}(\boldsymbol{\xi}) = \mathbf{q}^T(\boldsymbol{\xi}) \mathbf{a}^\alpha \cdot \mathbf{d}_N^{\beta}$$

$$M^{\alpha\beta}(\boldsymbol{\xi}) = \mathbf{q}^T(\boldsymbol{\xi}) \mathbf{a}_3 \cdot \mathbf{d}_M^{\alpha\beta}$$

$$\mathbf{q}(\boldsymbol{\xi}) = \{1, \xi^1, \xi^2\}^T$$



● : Meshfree nodes

● : Integration points for $\bar{g}_{\alpha I}, \bar{g}_{\alpha I}, \bar{g}_{\alpha\beta I}, \bar{g}_{\alpha\beta I}, \bar{K}, \bar{K}, \bar{f}, \bar{f}, \bar{f}$

✦ : Integration points for G, K

应力应变在背景积分域内求解

准变分一致型伽辽金无网格法

位移离散 ✓ 保证一致性

$$u^h(\xi) = \sum_{I=1}^{n_p} \Psi_I(\xi) d_I$$

$$p(\xi) = \{1, \xi^1, \xi^2, (\xi^1)^2, \xi^1 \xi^2, (\xi^2)^2\}^T$$

应变与应力离散 ✓ 保证变分一致性

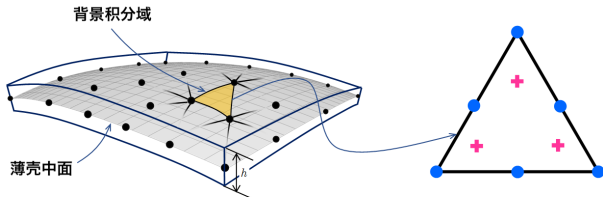
$$\varepsilon_{\alpha\beta}(\xi) = q^T(\xi) \frac{1}{2} (a_\alpha \cdot d_\beta^\varepsilon + a_\beta \cdot d_\alpha^\varepsilon)$$

$$\kappa_{\alpha\beta}(\xi) = q^T(\xi) a_3 \cdot d_{\alpha\beta}^\kappa$$

$$N^{\alpha\beta}(\xi) = q^T(\xi) a^\alpha \cdot d_N^\beta$$

$$M^{\alpha\beta}(\xi) = q^T(\xi) a_3 \cdot d_M^{\alpha\beta}$$

$$q(\xi) = \{1, \xi^1, \xi^2\}^T$$



●: Meshfree nodes

●: Integration points for $\bar{g}_{\alpha I}, \bar{g}_{\alpha I}, \bar{g}_{\alpha\beta I}, \bar{g}_{\alpha\beta I}, \bar{K}, \bar{K}, f, \tilde{f}, \bar{f}$

✦: Integration points for G, K

离散控制方程

$$(K + \tilde{K} + \bar{K})d = f + \tilde{f} + \bar{f}$$

采用低阶高斯积分采样点组装

与 Nitsche 法对比

一致项 ✓ 形函数二阶及以上导数采用光滑梯度代替，提高计算效率

HW 法:

$$\begin{aligned}\tilde{K}_{IJ} = & - \int_{\Gamma_v} (\Psi_I \tilde{\mathbf{T}}_{NJ}(\Psi_{J,\alpha}) + \tilde{\mathbf{T}}_{NI}(\Psi_{I,\alpha}) \Psi_J) d\Gamma \\ & + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma \mathbf{a}_3 \tilde{M}_{nnJ} + \mathbf{a}_3 \tilde{M}_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ & + ([[\Psi_I \mathbf{a}_3 \tilde{P}_J]] + [[\tilde{P}_I \mathbf{a}_3 \Psi_J]])_{\mathbf{x} \in C_v}\end{aligned}$$

Nitsche 法:

$$\begin{aligned}\mathbf{K}_{IJ}^c = & - \int_{\Gamma_v} (\Psi_I \mathbf{T}_{NJ}(\Psi_{J,\alpha\beta\gamma}) + \mathbf{T}_{NI}(\Psi_{I,\alpha\beta\gamma}) \Psi_J) d\Gamma \\ & + \int_{\Gamma_\theta} (\Psi_{I,\gamma} n^\gamma \mathbf{a}_3 \mathbf{M}_{nnJ} + \mathbf{a}_3 \mathbf{M}_{nnI} \Psi_{J,\gamma} n^\gamma) d\Gamma \\ & + ([[\Psi_I \mathbf{a}_3 \mathbf{P}_J]] + [[\mathbf{P}_I \mathbf{a}_3 \Psi_J]])_{\mathbf{x} \in C_v}\end{aligned}$$

稳定项 ✓ 自带稳定项，无需采用人工经验参数

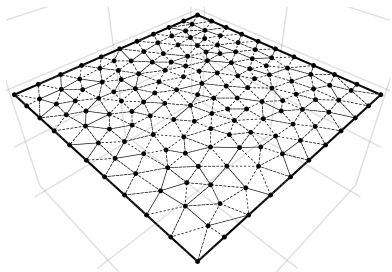
HW 法:

$$\bar{K}_{IJ} = - \int_{\Gamma_u} \bar{\mathbf{T}}_{MI} \Psi_J d\Gamma + \int_{\Gamma_\theta} \mathbf{a}_3 \bar{M}_{nnI} \Psi_{J,\gamma} n^\gamma d\Gamma + [[\bar{\mathbf{P}}_I \mathbf{a}_3 \Psi_J]]_{\mathbf{x} \in C_u}$$

Nitsche 法:

$$\mathbf{K}_{IJ}^s = \alpha_v \int_{\Gamma_v} \Psi_I \Psi_J d\Gamma + \alpha_\theta \int_{\Gamma_\theta} \Psi_{I,\eta} n^\eta \mathbf{a}_3 \mathbf{a}_3 n^\gamma \Psi_{J,\gamma} d\Gamma + \alpha_C [[\Psi_I \mathbf{a}_3 \mathbf{a}_3 \Psi_J]]_{\mathbf{x} \in C_u}$$

分片试验



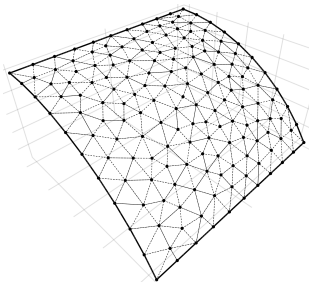
平面分片试验模型

$$\begin{cases} \mathbf{a}_1 = \{1.0, 0.0, 0.0\}^T \\ \mathbf{a}_2 = \{0.0, 1.0, 0.0\}^T \\ \mathbf{a}_3 = \{0.0, 0.0, 1.0\}^T \end{cases}$$

平面模型分片试验结果

	Linear patch test		Quadratic patch test	
	L_2 -Error	H_e -Error	L_2 -Error	H_e -Error
GI-Penalty	1.41E-04	4.62E-03	1.97E-03	7.17E-03
GI-Nitsche	1.73E-04	5.61E-03	1.85E-03	7.76E-03
RKGS1-Penalty	5.04E-09	1.02E-07	3.01E-09	3.41E-08
RKGS1-Nitsche	9.75E-12	8.98E-11	1.29E-12	1.06E-12
RKGS1-HR	6.15E-13	6.91E-12	7.51E-13	8.36E-12

分片试验



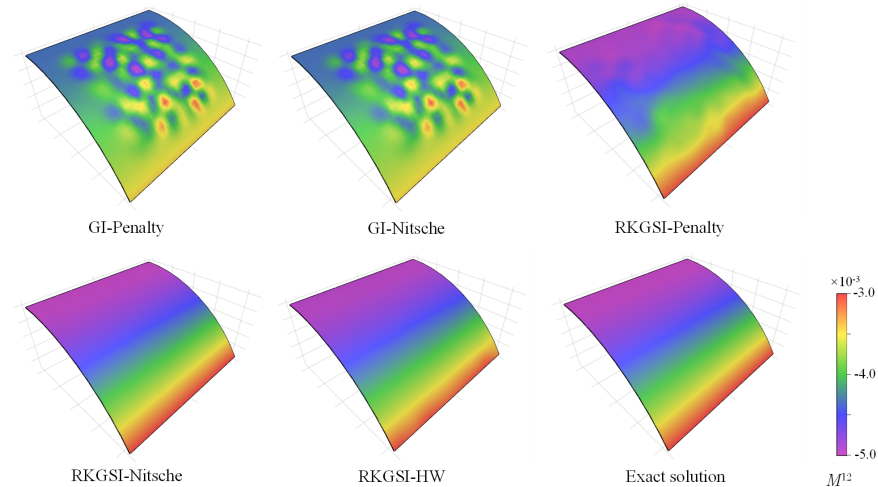
曲面分片试验模型

$$\begin{cases} \mathbf{a}_1 = \{\cos(\frac{\xi^1}{R}), 0.0, -\sin(\frac{\xi^1}{R})\}^T \\ \mathbf{a}_2 = \{0.0, 1.0, 0.0\}^T \\ \mathbf{a}_3 = \{\sin(\frac{\xi^1}{R}), 0.0, \cos(\frac{\xi^1}{R})\}^T \end{cases}$$

曲面模型分片试验结果

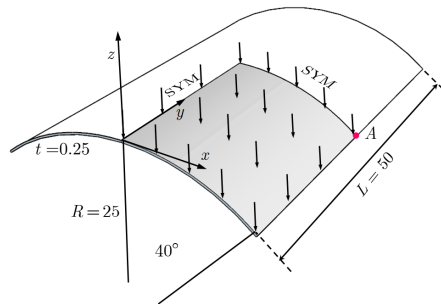
	Linear patch test		Quadratic patch test	
	L_2 -Error	H_e -Error	L_2 -Error	H_e -Error
GI-Penalty	1.75E-04	4.50E-03	1.08E-03	5.83E-03
GI-Nitsche	1.77E-04	5.36E-03	1.07E-03	6.33E-03
RKGSi-Penalty	8.59E-05	9.11E-04	4.28E-04	2.08E-03
RKGSi-Nitsche	1.27E-05	5.32E-05	1.88E-05	5.6E-04
RKGSi-HR	1.43E-05	1.60E-04	2.93E-05	2.85E-04

分片试验



曲壳分片试验 M^{12} 云图

Scordelis-Lo 屋顶问题



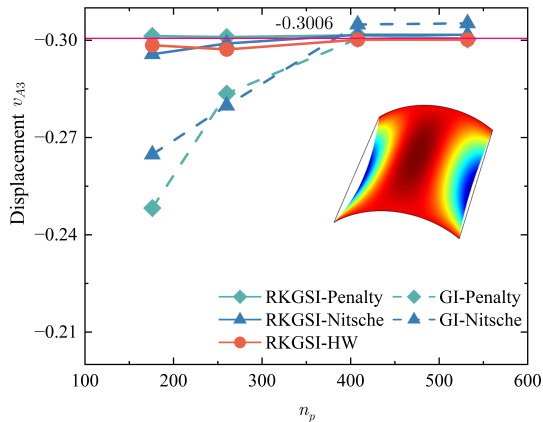
Scordelis-Lo 屋顶问题模型

杨氏模量 $E = 4.32 \times 10^8$

泊松比 $\nu = 0.0$

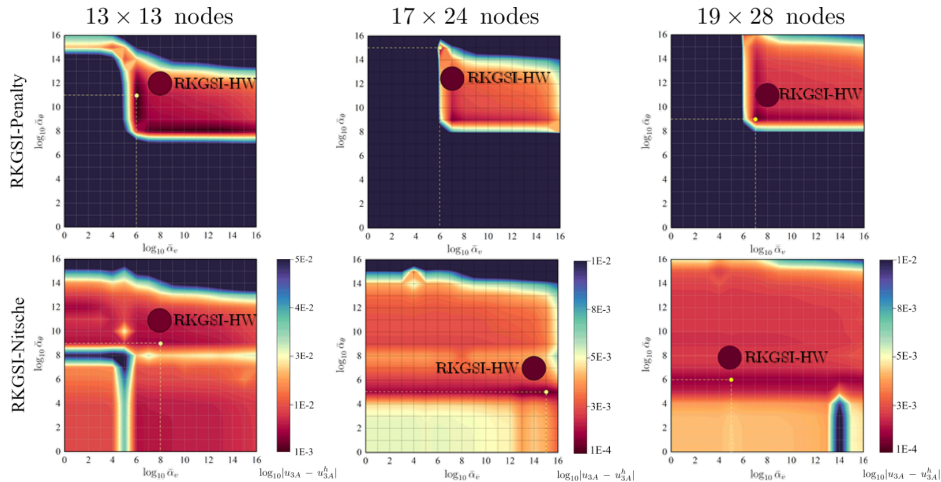
厚度 $h = 0.25$

荷载 $q = 90.0$



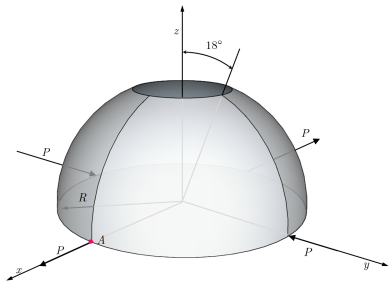
位移收敛性分析

Scordelis-Lo 屋顶问题



人工经验参数敏感性分析

夹支半球壳问题



夹支球壳问题模型

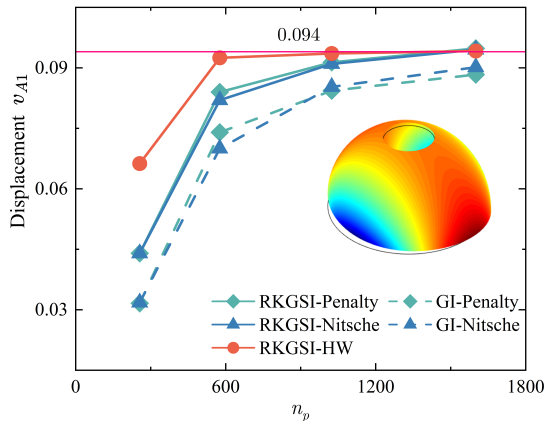
杨氏模量 $E = 6.825 \times 10^7$

泊松比 $\nu = 0.3$

半径 $R = 10$

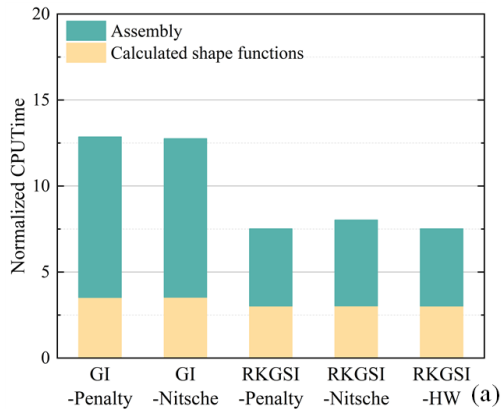
厚度 $h = 0.04$

荷载 $P = \pm 2$

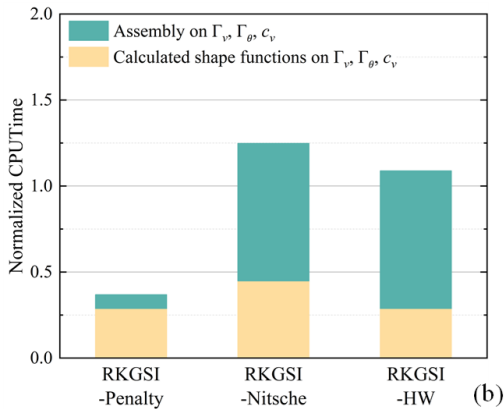


位移收敛性分析

夹支半球壳问题



全域效率对比



本质边界效率对比

结论

- 基于 Hu-Washizu 变分原理提出了薄壳结构分析的准变分一致伽辽金无网格法。
- 该方法在平面模型中可满足变分一致性，在曲面模型中可满足多项式阶次的变分一致性条件。
- 该方法采用光滑梯度替代传统形函数二阶及以上形函数导数，提高计算效率。
- 该方法包含本质边界条件施加方案，施加过程无需人工经验参数即可保证计算精度。

相关工作:

1. Wu, J., Xu, Y., Xu, B., Basha, S.H., 2024. Quasi-consistent efficient meshfree thin shell formulation with naturally stabilized enforced essential boundary conditions. *Engineering Analysis with Boundary Elements* 163, 641–655.
2. Wu, J., Wu, X., Zhao, Y., Wang, D., 2022. A consistent and efficient method for imposing meshfree essential boundary conditions via Hellinger-Reissner variational principle. *Chinese Journal of Theoretical and Applied Mechanics* 54, 3283–3296.
3. 吴俊超, 吴新瑜, 赵珧冰, 王东东, 2022. 基于赫林格-赖斯纳变分原理的一致高效无网格本质边界条件施加方法. *力学学报* 54, 3283–3296.



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谢谢!